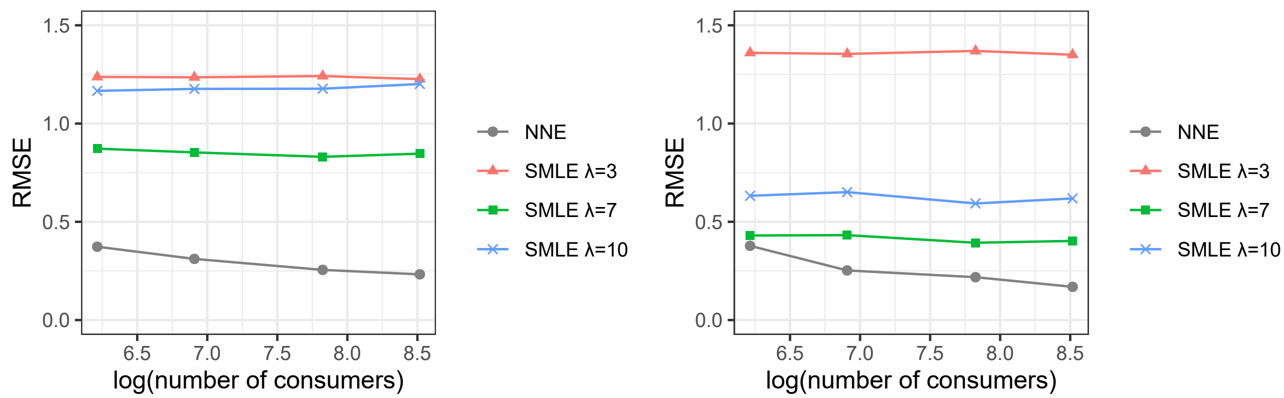

Notes: All reported numbers are averaged across 100 Monte Carlo datasets. Within each plot, L^* and R are chosen so that NNE and SMLE have about the same simulation burdens. In the left plot $L^* = 1e4$ and $R = 15$. In the right plot $L^* = 4e4$ and $R = 50$.

Figure 9: Estimation of Search Model with Different Data Sizes

made to have about the same simulation burdens. For SMLE we include three smoothing factors: $\lambda = 3, 7$, and 10. As before, we include $\lambda = 7$ because a grid search finds it optimal for SMLE at $n = 1000$ (Figure 9 here suggests that $\lambda = 7$ is also optimal for the other values of n). We again note that finding this optimal λ not only is computationally expensive but also uses the knowledge of the true search model parameter.

The most important observation from Figure 9 is that the RMSE of NNE decreases with n faster than SMLE. In other words, NNE can capitalize on larger data better. Intuitively, this is because the smoothing required by SMLE introduces bias in estimates (as we have seen in Section 4.3). This bias is not a finite-sample bias and tends not to decrease with n. To reduce this bias and subsequently RMSE, one needs to increase R and use a larger λ (the optimal λ increases with R). However, a larger R also implies a higher computational cost.

5 Conclusion

We study a novel approach that leverages machine learning to estimate the parameters of (structural) econometric models. One uses the structural econometric model to generate datasets. These generated datasets, together with the parameter values under which they are generated, can be used to train a machine learning model to "recognize" parameter values from datasets. As such, the approach offers a bridge to connect the existing machine learning techniques and the current structural econometric models.

NNE has its advantages and boundaries. We find that it has a well-defined and meaningful limit, is robust to redundant moments, and achieves good estimation accuracy at light computational costs in suitable applications. The applications that can benefit the most from NNE are where SMLE or SMM would require large numbers of simulations. NNE is unlikely to show gains in applications where the main estimation burden is not simulations. One example is where closed-form likelihood